The main thing is not to oversalt — Solution

W21 (2021) — English translation

A1 ?? pts

Measure the resistance of the limiting resistor R_1 with an ohmmeter (see equipment). The approximate value of 10 ohms is indicated below, but use the more accurate value you obtained in your calculations.

Let's measure the resistance of the limiting resistor $R_1 = 10.1\ \Omega$ with a multimeter in ohmmeter mode.

A2 7.50 pts

Experimentally confirm the validity of the proposed model. To do this, study the current-voltage characteristics of the solution at 5 different distances between the electrodes.

We assemble a measuring circuit consisting of a battery connected in series, a control unit, and a syringe with a salt solution. We connect a voltmeter to the electrodes of the syringe. We connect the second voltmeter in parallel with the reference resistance R_1 to measure the current in the circuit.

The entire set of processes on the cathode and anode is proposed to be represented in the form of a series circuit consisting of an external applied voltage U, the resistance of the electrolyte volume R, the contact EMF U_c, and the contact resistance r_c (Fig. 1). It is assumed that the value R is directly proportional to the resistivity of the electrolyte ρ and the distance between the electrodes L, and inversely proportional to the cross-sectional area of the electrolytic bath S.

The theoretical current-voltage characteristic of the electrolytic bath within the framework of this model has the form:

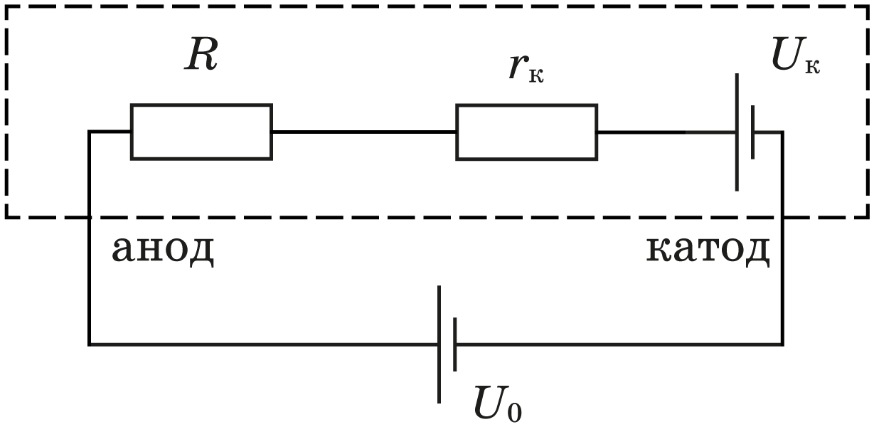
$$U = U_c + I\left(r_c + \rho\frac{L}{S}\right)$$

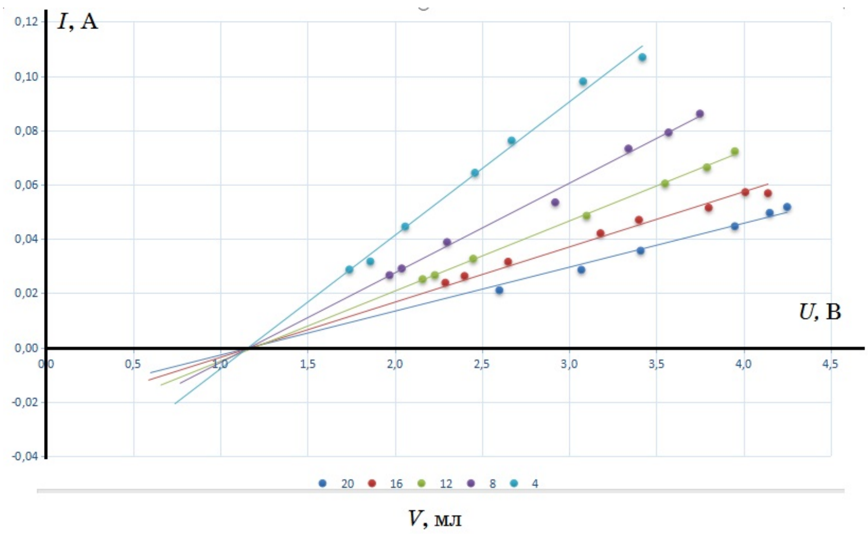
We obtain the current-voltage characteristic of the solution. The tables below show the results of measuring the current-voltage characteristics of the solution at 5 different volumes of liquid in the syringe V (and the corresponding distances L between the electrodes).

$V = 4$ **ml,** $L = 1.3$ **cm**			$V = 8$ **ml,** $L = 2.6$ **cm**			$V = 12$ **ml,** $L = 3.9$ **cm**		
U_{10} (mV)	U (mV)	I (mA)	U_{10} (mV)	U (mV)	I (mA)	U_{10} (mV)	U (mV)	I (mA)
290	1740	28.7	870	3750	86.1	254	2160	25.1
320	1860	31.7	800	3570	79.2	269	2230	26.6
450	2060	44.6	740	3340	73.3	330	2450	32.7
650	2460	64.4	540	2920	53.5	490	3100	48.5
770	2670	76.2	392	2300	38.8	610	3550	60.4
990	3080	98.0	294	2040	29.1	670	3790	66.3
1080	3420	106.9	269	1970	26.6	730	3950	72.3

$V = 16$ **ml,** $L = 5.2$ **cm**			$V = 20$ **ml,** $L = 6.5$ **cm**		
U_{10} (mV)	U (mV)	I (mA)	U_{10} (mV)	U (mV)	I (mA)
574	4140	56.8	213	2600	21.1
578	4010	57.2	289	3070	28.6
520	3800	51.5	360	3410	35.6
475	3400	47.0	451	3950	44.7
425	3180	42.1	501	4150	49.6
319	2650	31.6	523	4250	51.8
266	2400	26.3			

We plot the current-voltage characteristics for all experiments:





A3 1.00 pts

Estimate the magnitude of the contact potential difference U_c.

Using the graph, we determine:

1. From the point of intersection of the graphs and the voltage axis, the value of the contact potential difference is $U_c = 1.2$ V.
2. From the slope (see the theoretical formula), we can find the resistance of the electrolyte volume together with the contact resistance R_{tot} for the 5 values of L.

A4 2.50 pts

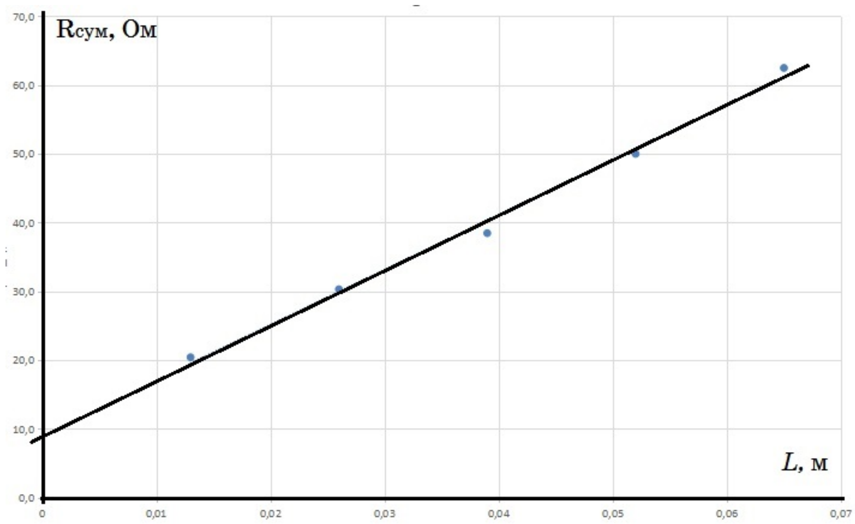
Determine the value of the contact resistance r_c.

The results of processing the experimental data for the five values of L are presented in the table and figure below.

No.	L, m	$1/R_{\mathbf{tot}}$, Ω^{-1}	$R_{\mathbf{tot}}$, Ω
1	0.065	0.016	62.5
2	0.052	0.020	50.0
3	0.039	0.026	38.5
4	0.026	0.033	30.3
5	0.013	0.049	20.4

From the graph, we find the y-intercept corresponding to $L = 0$, which gives us the contact resistance:

$$r_c = R_{\text{tot}}(0) = 9.2\ \Omega$$



A5 1.00 pts

Calculate the specific conductivity σ of a three percent solution of sodium chloride in water.

The cross-sectional area of the piston is $S = 3.1 \cdot 10^{-4}\ \text{m}^2$ (this can be found by measuring the diameter of the piston or by the ratio of volume to length). Within the framework of the used model:

$$\frac{\Delta R_{\text{tot}}}{\Delta L} = \frac{\rho}{S}$$

From the slope of the straight line $R_{\text{tot}}(L)$, we find the resistivity of the salt solution used:

$$\rho = 0.24\ \Omega \cdot \text{m}$$

Therefore, the specific conductivity is:

$$\sigma = \frac{1}{\rho} = 4.2\ (\Omega \cdot \text{m})^{-1}$$

A6 3.00 pts

Calculate the mobility μ of sodium and chlorine ions in this solution.

The specific conductivity of the electrolyte is given by:

$$\sigma = \frac{1}{\rho} = q_1 n_1 \mu_1 + q_2 n_2 \mu_2$$

where q_1 and q_2 are the absolute electrical charges of the positive and negative ions, respectively (for sodium and chlorine ions $q_1 = q_2 = e$, where e is the elementary charge). n_1 and n_2 are their concentrations (in our case they are the same and equal to n), and μ_1 and μ_2 are the mobilities of the negative and positive ions in solution, which, according to the conditions of the problem, are approximately equal to each other ($\mu_1 = \mu_2 = \mu$). Thus, the equation simplifies to:

$$\mu = \frac{\sigma}{2en}$$

We calculate the concentration of sodium (or chlorine) ions in a three percent solution of table salt (NaCl):

$$n = \frac{N}{V} = \frac{m_{\mathrm{NaCl}} N_A \rho_{\mathrm{sol}}}{M_{\mathrm{NaCl}} m_{\mathrm{sol}}} = 0.03 \frac{N_A \rho_{\mathrm{sol}}}{M_{\mathrm{NaCl}}} = 3.2 \cdot 10^{26}\ \mathrm{m}^{-3}$$

Now, we calculate the mobility:

$$\mu = \frac{\sigma}{2en} = 4.0 \cdot 10^{-8}\ \frac{\mathrm{m}^2}{\mathrm{V} \cdot \mathrm{s}}$$

which is in good agreement with established literature data.